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COURSE CODE : CSA0613

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**COURSE NAME :Design and
Analysis of Algorithms**

C01-ASSESSMENT TOOL 03 -TECHNICAL PRESENTATION

TITLE : Divide and Conquer Algorithm Efficiency Evaluation Analyze a divide-and-conquer algorithm with recurrence $T(n) = 2T(n/2) + n \log n$.



PROJECT DESCRIPTION

This project focuses on evaluating the efficiency of a Divide and Conquer algorithm using the recurrence relation $T(n) = 2T(n/2) + n \log n$. The algorithm works by dividing a large problem into two equal subproblems, solving each subproblem recursively, and then combining their results. The recurrence relation is analyzed using the Master Theorem to determine the overall time complexity, while a recursion tree is used to visualize the recursive breakdown and the work performed at each level. The project also includes the algorithm, pseudocode, architectural diagram, Python implementation, and output analysis to demonstrate how the recurrence works in practice. The obtained result shows that the algorithm has a time complexity of $\Theta(n \log^2 n)$, making it more efficient than quadratic-time algorithms for large input sizes. This project helps in understanding recursion, recurrence relations, Divide and Conquer strategy, and the application of the Master Theorem in analyzing algorithm performance.